

BALL BALANCING PID CONTROLLED ROBOT

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Abstract—Balancing a ball on a beam is a classic benchmark problem in control systems, demonstrating the challenges of stabilizing an unstable system. This paper presents the design, implementation, and testing of a low-cost, one-dimensional ball-balancing robot. The system is built using an Arduino UNO microcontroller, which processes real-time position data from an HC-SR04 ultrasonic sensor. A servomechanism acts as the actuator, tilting the platform to control the ball's position. The core of the system is a Proportional-Integral-Derivative (PID) control algorithm, which continuously calculates the error between the ball's current position and a desired setpoint. The physical prototype was constructed and tested, demonstrating the system's ability to successfully balance the ball and reject external disturbances, quickly returning the ball to the setpoint. This project provides an effective, hands-on implementation of a closed-loop PID control system.

Keywords—Ball balancing robot, PID controller, Arduino, Control Systems, HC-SR04, Ultrasonic Sensor, Servomechanism

I. INTRODUCTION

Stabilizing an unstable system is a fundamental and classic challenge in control engineering. The ball balancing problem is a benchmark experiment used to demonstrate and test control theory in practice. The objective is to design a system that can automatically balance a ball at a desired position on a moving platform by tilting the platform, a task that is inherently unstable as the ball will naturally roll off. The principles demonstrated in this project have significant real-world applications. They are foundational to the operation of self balancing robots , stabilization systems in ships and aircraft, precise material handling in automated manufacturing, and advanced robotics. Many complex control strategies have been applied to this problem. However, the Proportional-Integral-Derivative (PID) controller remains one of the most effective and widely used methods due to its robustness, simplicity, and excellent performance[1]. This paper presents the design, construction, and implementation of a low-cost, one-dimensional ball balancing robot. The system utilizes an Arduino UNO microcontroller as the main processor, an HC-SR04 ultrasonic sensor for real-time position sensing, and a servo motor as the actuator. A PID control algorithm is implemented in software to process the sensor data and compute the necessary adjustments to the platform's angle, creating a closed-loop feedback system. This report is organized as follows: Section II details the system model, hardware components, and circuit design. Section III explains the PID control algorithm and its implementation. Section IV presents the experimental results and a discussion of the system's performance. Finally, Section V

concludes the paper and suggests potential directions for future work

II. SYSTEM MODEL AND METHODS

The project is implemented as a closed-loop feedback control system. The primary goal is to maintain a ball at a specific setpoint by continuously measuring its position and adjusting the platform's angle to correct any error.

A. Hardware Components

The system is constructed from three primary hardware components:

1) Arduino UNO: The Arduino UNO, based on the ATmega328P microcontroller, serves as the central processing unit[2]. It is responsible for reading the position data from the ultrasonic sensor, executing the PID control algorithm to calculate the required correction, and generating a pulse-width modulation (PWM) signal for the servo motor.

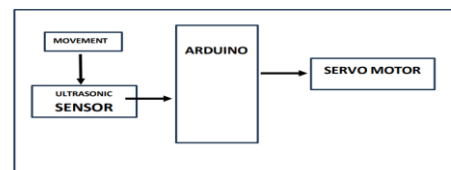
2) HC-SR04 Ultrasonic Sensor: To detect the ball's position, an HC-SR04 ultrasonic sensor is used[3]. This sensor provides non-contact distance measurement by emitting ultrasonic pulses and measuring the time-of-flight for the echo to return. This time is then converted into a distance, representing the ball's current position.

3) Servo Motor: A standard servo motor (e.g., SG90) is used as the actuator. It receives the PWM control signal from the Arduino, which precisely adjusts the motor's angle. This angle is mechanically linked to the platform, allowing the system to tilt the platform and move the ball.

B. System Design and Implementation

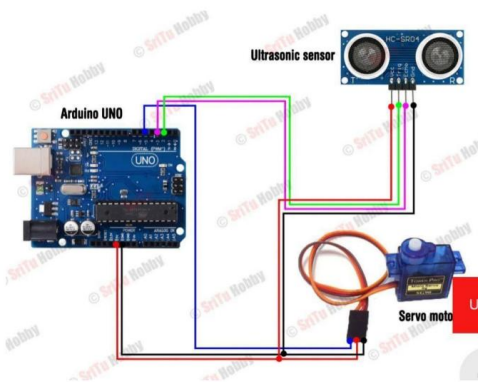
The overall system architecture is shown in the block diagram in Fig. 1. The ultrasonic sensor provides the position input, which is fed to the Arduino (the controller). The Arduino then computes a control signal and sends it to the servo motor (the actuator) to produce a physical movement, which in turn affects the ball's position. Fig. 1. System block diagram.

Block Diagram:



The complete wiring and electrical connections are detailed in the circuit diagram in Fig. 2. The sensor and servo are powered by the Arduino's 5V pin. The sensor's Trigger and Echo pins, as well as the servo's signal pin, are connected to the Arduino's digital I/O pins for data transfer and control. Fig. 2. System circuit diagram.

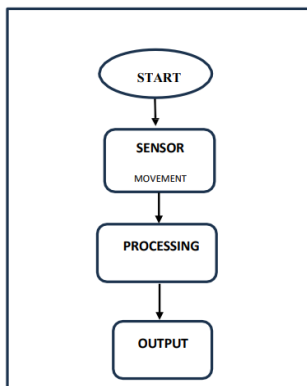
CIRCUIT DIAGRAM:



C. System Workflow

The operational logic of the system follows a continuous loop, as shown in the flowchart in Fig. 3. The process begins by reading the current ball position from the sensor. This value is then compared to the desired setpoint to calculate an error. The error is fed into the PID algorithm for processing, which generates a new control signal. This signal is sent to the servo, which adjusts the platform, and the entire loop repeats Fig. 3. System control logic flowchart.

Flow Chart:



D. Control Algorithm

The system's stability is achieved through a Proportional-Integral-Derivative (PID) controller. This algorithm calculates a control signal by operating on the error $e(t)$, which is the difference between the setpoint and the measured position. The PID output is the sum of three terms:

- **1) Proportional (P) Control:** The P-term provides an output proportional to the current error ($K_p \cdot e(t)$). It is responsible for the primary corrective action. A larger proportional constant (K_p) results in a faster response but can lead to instability and overshoot.

- **2) Integral (I) Control:** The I-term integrates the error over time ($K_i \int e(t) dt$). This term is designed to eliminate steady-state error, which is the long-term, persistent error that the P-controller may fail to correct on its own.
- **3) Derivative (D) Control:** The D-term differentiates the error with respect to time ($K_d \cdot de(t)/dt$). This term predicts the future behavior of the error and provides a damping effect, which helps to reduce overshoot and oscillations.

The final control output sent to the servo is the sum of these three terms. The constants K_p , K_i , and K_d are the tuning parameters that must be carefully adjusted to achieve a stable, fast, and accurate response.

III. RESULTS AND DISCUSSION

After constructing the system and implementing the PID algorithm, the controller's parameters (K_p , K_i , K_d) were manually tuned through an iterative process of trial and error. The goal was to achieve a response that was both fast and stable, minimizing overshoot and oscillation.

A. Performance Analysis

The system's performance was evaluated by placing the ball on the platform and observing the controller's ability to achieve two key objectives:

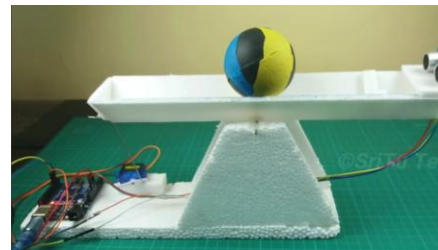
Setpoint Tracking: Bringing the ball to the center setpoint from an arbitrary starting position.

Disturbance Rejection: Returning the ball to the setpoint after it was manually pushed or the platform was disturbed.

B. Experimental Results

The physical prototype successfully demonstrated stable balancing, as documented in the project video[4]. Fig. 4 shows a snapshot of the system in operation.

Fig. 4. Physical prototype balancing the ball.



The system's behavior, shows a stable response. When the ball is moved, the ultrasonic sensor correctly detects the change in position, and the PID controller immediately actuates the servo motor to tilt the platform in the opposite direction. The platform's angle is continuously adjusted to push the ball back toward the center. The system successfully rejects small and medium-sized disturbances, returning the ball to the setpoint with minimal oscillation.

C. Discussion

The successful implementation confirms the effectiveness of the PID control strategy for this type of unstable system. This project has several advantages and potential applications.

Applications: The technology, while basic, is a core component for more advanced systems. It can be scaled for precise material handling in industrial automation, used in warehouse robotics to transport sensitive goods, or even serve as a basis for stabilization in marine vehicles.

Limitations: The current system also has several confines. The reliance on a PID controller, while effective, requires careful manual tuning and high computational power for a fast response. The ultrasonic sensor, while low-cost, can be sensitive to external factors like temperature and pressure, which could affect the accuracy of its distance readings. Furthermore, this design is one dimensional, a more complex two dimensional balancing platform would require additional sensors and actuators.

IV. CONCLUSION

This paper successfully presented the complete design, implementation, and testing of a one-dimensional ball balancing robot. The system, built with an Arduino UNO, an HC-SR04 ultrasonic sensor, and a servo motor, demonstrates an effective real-world application of control system theory. By implementing a PID control algorithm, the system was able to stabilize the ball and maintain its position at a desired setpoint, effectively rejecting external disturbances. The project confirms that this combination of low-cost, readily available hardware can be used to create a robust closed-loop feedback system. While the current system is limited to one-dimensional balancing and relies on manual PID tuning, it serves as a strong foundation for future development. Future work could involve expanding the system to two dimensions (a ball on a plate), implementing an auto-tuning PID algorithm, or integrating a camera with computer vision for more precise position detection.

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